Day 8 Activity: Plot-matching with `penguins`

Thursday, May 28, 2026

Your Name Here

Setup

```r
library(tidyverse)
library(palmerpenguins)
```

Today's activity uses the `penguins` dataset (`palmerpenguins` package). Glimpse it before you start:

```r
glimpse(penguins)
```

```
Rows: 344
Columns: 8
$ species           <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
$ island            <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
$ bill_length_mm    <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
$ bill_depth_mm     <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
$ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
$ body_mass_g       <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
$ sex               <fct> male, female, female, NA, female, male, female, male~
$ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

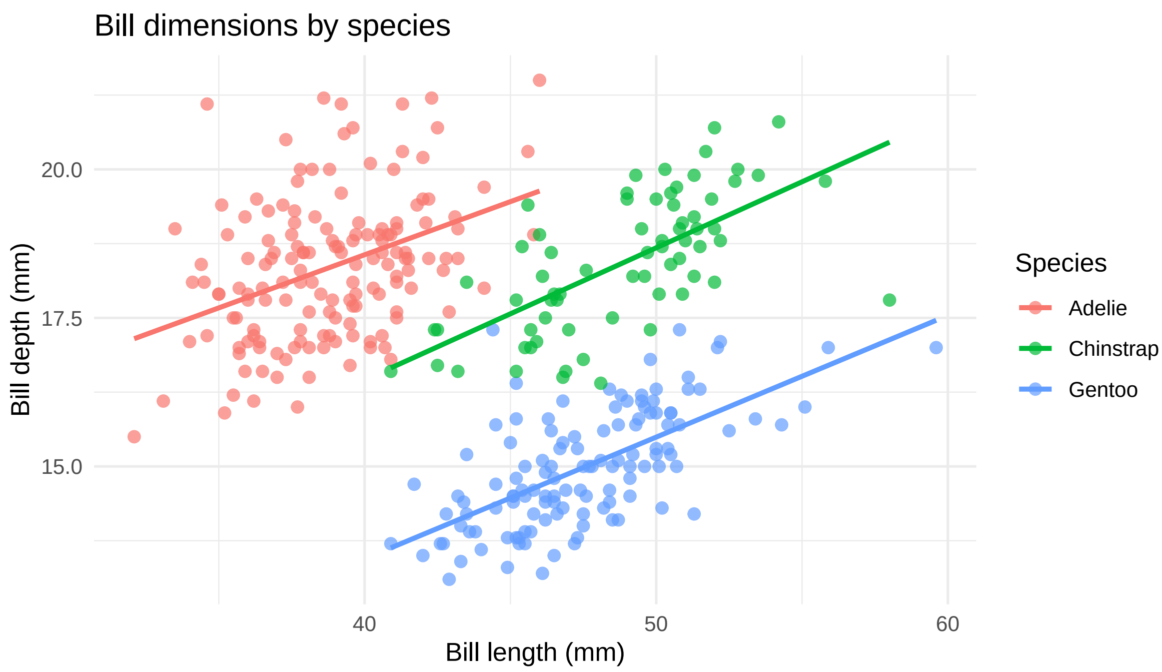
Instructions

Below are **four target plots**, generated for you in advance. Your job is to **recreate each one as closely as you can** using `ggplot2`, then write **one sentence** about what each plot reveals about the penguins.

You won't get perfect matches. Themes, exact colors, and tiny details may differ. Aim to match:

- The frame (which variables are on x and y)
- The glyph (which geom)
- The aesthetic mappings (color, shape, facets, etc.)
- Any axis transformations or labels
- Anything else that's clearly *intentional* in the target

Target 1 — Bills, with species and a smoother

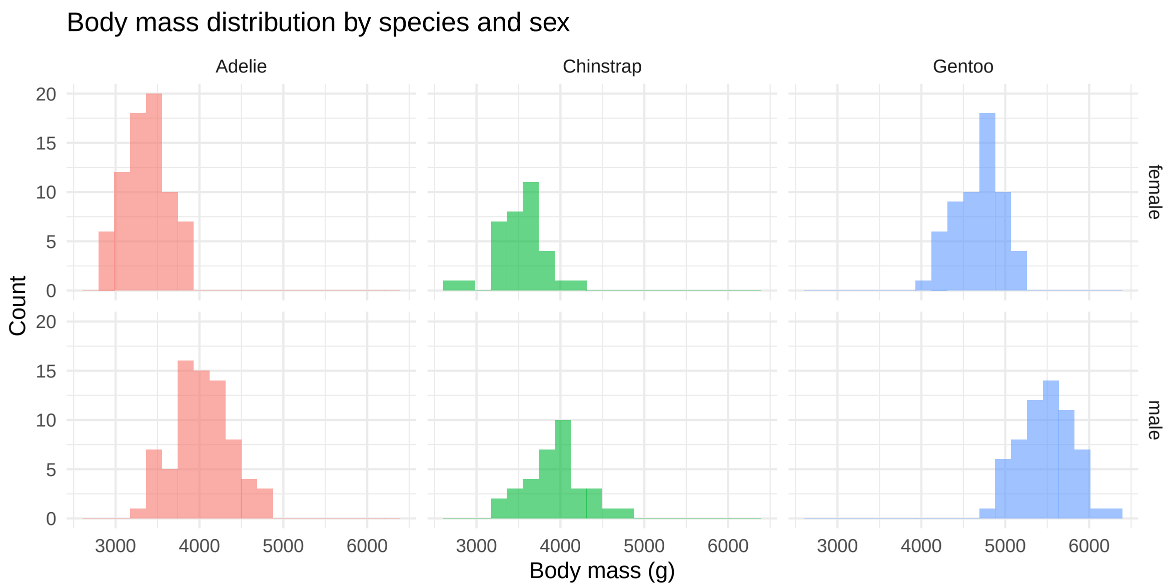



```
# Your code:
```

What does this plot reveal?

[Write one sentence.]

Target 2 — Body mass distribution by species and sex



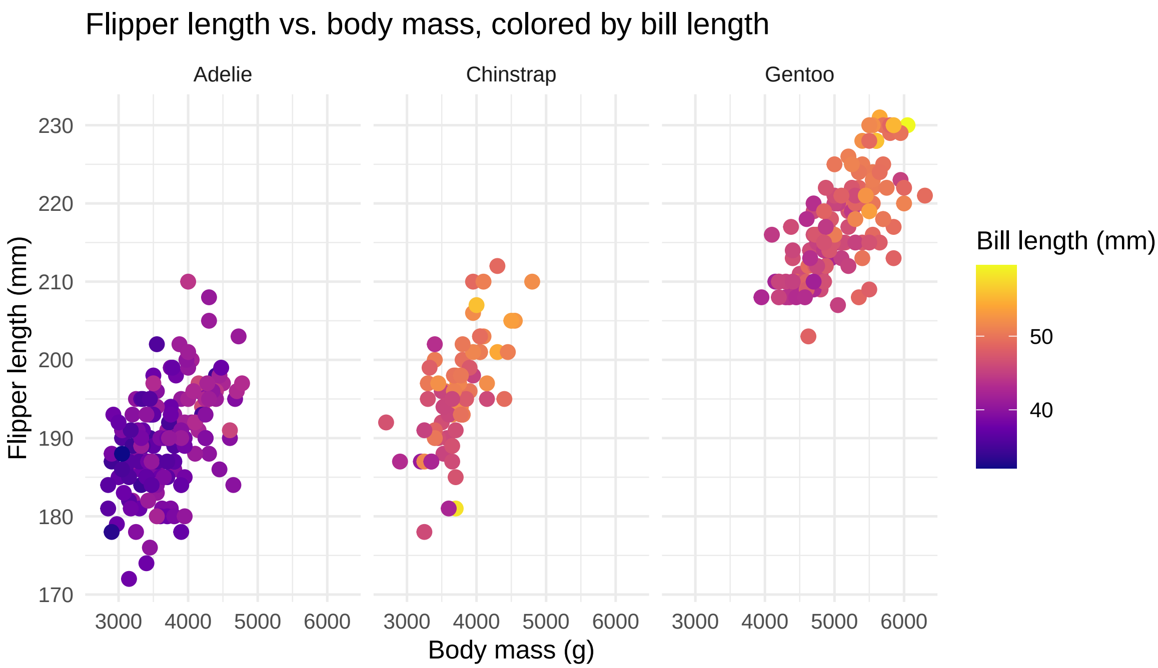

```
# Your code:
```

Hint: the legend is hidden because the fill is redundant with the facet columns. You can use `show.legend = FALSE` on the geom, or `theme(legend.position = "none")` to do this.

What does this plot reveal?

[Write one sentence.]

Target 3 — Flipper length vs. body mass, with continuous color



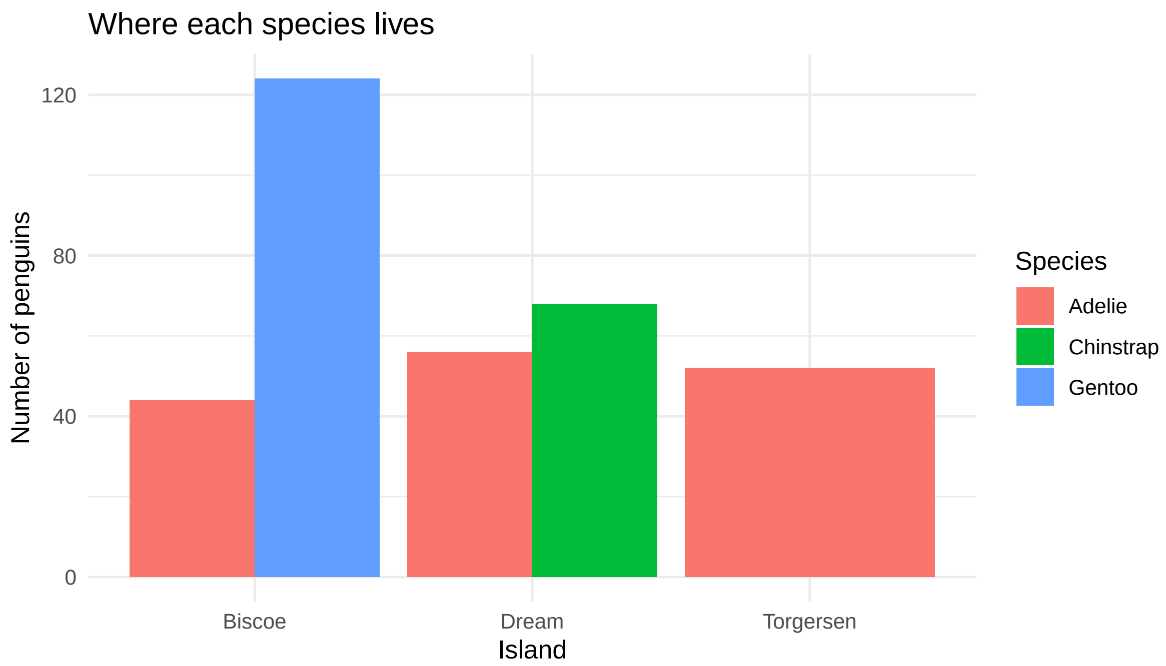

```
# Your code:
```

*Hint: this plot encodes **four** variables. The continuous color scale is from the viridis package.

What does this plot reveal?

[Write one sentence.]

Target 4 — Counts of penguins by island and species




```
# Your code:
```

Hint: pay attention to the position adjustment. Stacked, dodged, and filled bars all look different.

What does this plot reveal?

[Write one sentence.]

Wrap-up

Once you've matched all four, answer the following question:

Which target was hardest? Was it the one with the most variables, or one with a subtler design choice?

[Write 1–2 sentences.]

Code appendix

```
library(tidyverse)
library(palmerpenguins)
glimpse(penguins)
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm, color = species)) +
  geom_point(size = 2, alpha = 0.7, na.rm = TRUE) +
  geom_smooth(method = "lm", se = FALSE, na.rm = TRUE) +
  labs(
    x = "Bill length (mm)",
    y = "Bill depth (mm)",
    title = "Bill dimensions by species",
    color = "Species"
  ) +
  theme_minimal()
# Your code:

penguins |>
  filter(!is.na(sex)) |>
  ggplot(aes(x = body_mass_g, fill = species)) +
  geom_histogram(bins = 20, position = "identity", alpha = 0.6) +
  facet_grid(sex ~ species) +
  labs(
    x = "Body mass (g)",
    y = "Count",
    title = "Body mass distribution by species and sex",
    fill = "Species"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
# Your code:

ggplot(penguins, aes(x = body_mass_g, y = flipper_length_mm, color = bill_length_mm)) +
  geom_point(size = 2.5, na.rm = TRUE) +
  scale_color_viridis_c(option = "plasma") +
  facet_wrap(~species) +
  labs(
    x = "Body mass (g)",
    y = "Flipper length (mm)",
    title = "Flipper length vs. body mass, colored by bill length",
    color = "Bill length (mm)"
```

```r
  ) +
  theme_minimal()
# Your code:

ggplot(penguins, aes(x = island, fill = species)) +
  geom_bar(position = "dodge") +
  labs(
    x = "Island",
    y = "Number of penguins",
    title = "Where each species lives",
    fill = "Species"
  ) +
  theme_minimal()
# Your code:
```